

5. 32-Lecture Detailed Course Plan

MODULE 1 — Foundations of Bioinformatics & Biological Data

Lecture 1 — Introduction to Bioinformatics

Topics

- What is bioinformatics?
- Computational biology vs bioinformatics
- Why biological data is different from conventional data
- Role of Data Science in biology
- Applications in:
 - healthcare
 - drug discovery
 - agriculture
 - infectious diseases
 - precision medicine
 - biotechnology

Data Science connection

Biological Question → Data → Processing → Modeling → Biological Interpretation

Practical

Explore NCBI, UniProt and Ensembl.

Lecture 2 — Molecular Biology for Data Scientists

Topics

- DNA
- RNA
- proteins
- genes
- chromosomes
- genome
- transcriptome
- proteome
- central dogma
- transcription
- translation
- codons
- mutations

Data Science connection

DNA → sequence data

RNA → expression data

Protein → sequence/structure/function data

Practical

Use Python to calculate:

- GC content
- sequence length
- nucleotide frequency
- reverse complement

Lecture 3 — Biological Databases & Data Retrieval

Topics

- Primary vs secondary databases
- NCBI
- GenBank
- SRA
- GEO
- Ensembl
- UCSC
- UniProt
- PDB
- ClinVar
- gnomAD
- AlphaFold DB

Practical

Retrieve a gene sequence from NCBI and inspect its annotation.

Lecture 4 — Biological Data Formats & Linux

Topics

- FASTA
- FASTQ
- SAM
- BAM
- VCF
- BED
- GFF/GTF
- TSV/CSV
- metadata

Linux

- grep
- awk
- sed
- head
- tail
- sort
- cut
- pipes
- redirection

Practical

Analyze a FASTQ file from the command line.

MODULE 2 — Sequence Bioinformatics

Lecture 5 — Sequence Alignment

Topics

- Why alignment?
- Pairwise alignment
- Global alignment
- Local alignment
- Needleman-Wunsch
- Smith-Waterman
- scoring matrices
- gap penalties

Data Science connection

Optimization problem + dynamic programming.

Practical

Perform sequence alignment using Python/Biopython.

Lecture 6 — BLAST & Similarity Search

Topics

- BLAST concept
- BLASTn
- BLASTp
- BLASTx
- E-value
- bit score
- identity

- similarity
- coverage

Practical

Run BLAST against NCBI.

Student task

Given an unknown protein sequence:

Identify the most likely protein family.

Lecture 7 — Multiple Sequence Alignment & Motif Discovery

Topics

- Multiple sequence alignment
- conserved regions
- motifs
- domains
- MAFFT
- MUSCLE
- HMMER
- position-specific scoring

Practical

Analyze a protein family using MAFFT.

Lecture 8 — Phylogenetics & Evolutionary Data Analysis

Topics

- Evolutionary relationships
- distance matrices
- neighbor joining
- maximum likelihood
- phylogenetic trees
- bootstrap
- molecular evolution

Practical

Generate a phylogenetic tree.

Data Science connection

Clustering + distance metrics + statistical confidence.

MODULE 3 — NGS & GENOMIC DATA SCIENCE

Lecture 9 — Next-Generation Sequencing

Topics

- Sanger sequencing
- Illumina
- short-read sequencing
- long-read sequencing
- Oxford Nanopore
- PacBio HiFi
- sequencing coverage
- read quality

Comparison

Technology	Major characteristic
Illumina	highly accurate short reads
PacBio HiFi	accurate long reads
Oxford Nanopore	very long reads + real-time sequencing

Lecture 10 — NGS Quality Control

Topics

- sequencing errors
- Phred score
- read quality
- adapter contamination
- GC bias
- duplication
- quality trimming

Tools

- FastQC
- MultiQC
- Fastp

Practical

Perform QC on real FASTQ data.

Lecture 11 — Read Alignment & Genome Mapping

Topics

- reference genome
- mapping
- alignment
- indexing
- MAPQ
- SAM/BAM
- BWA
- Bowtie2
- minimap2

Practical

Map sequencing reads to a reference genome.

Lecture 12 — Genome Assembly & Annotation

Topics

- reference-based assembly
- de novo assembly
- overlap-layout-consensus
- de Bruijn graphs
- long-read assembly
- genome annotation
- gene prediction

Modern tools

- SPAdes
- Flye
- hifiasm
- QUAST
- Prokka
- Bakta

Practical

Assemble a small microbial genome.

MODULE 4 — VARIANTS & PRECISION GENOMICS

Lecture 13 — Variant Calling

Topics

- SNP
- SNV
- insertion
- deletion
- structural variant
- CNV
- germline vs somatic variants

Pipeline

FASTQ → QC → Alignment → BAM → Variant Calling → VCF

Tools

- GATK
- bcftools
- DeepVariant

Lecture 14 — Variant Annotation & Interpretation

Topics

- VCF
- functional annotation
- ClinVar
- gnomAD
- dbSNP
- VEP
- SnpEff
- pathogenic vs benign variants

Practical

Annotate variants and interpret their possible biological consequences.

Lecture 15 — Structural Variants & Long-Read Bioinformatics

Topics

- large insertions

- deletions
- inversions
- translocations
- repeat expansions
- long-read alignment
- structural variant detection

Tools

- minimap2
- Sniffles
- cuteSV
- pbsv

Practical

Compare short-read and long-read variant detection.

Lecture 16 — Pangenomics & Graph Genomes

Topics

- limitations of single reference genomes
- population variation
- pangenome
- graph genomes
- haplotypes
- genome graphs
- variation-aware alignment

Tools/concepts

- VG
- Minigraph
- Minigraph-Cactus
- Giraffe

Assessment

Students analyze the difference between:

Linear Reference Genome

and

Graph/Pangenome Reference

MODULE 5 — TRANSCRIPTOMICS

Lecture 17 — RNA-seq Data Analysis

Topics

- transcriptome
- RNA-seq workflow
- read alignment
- transcript quantification
- gene counts
- TPM
- FPKM
- counts

Tools

- STAR
- Salmon
- kallisto

Practical

Generate gene-expression matrix.

Lecture 18 — Differential Gene Expression

Topics

- biological vs technical variation
- null hypothesis
- p-value
- fold change
- log2 fold change
- FDR
- multiple testing
- Benjamini-Hochberg

Tools

- DESeq2
- edgeR
- limma

Practical

Identify differentially expressed genes.

Lecture 19 — Functional Enrichment & Pathway Analysis

Topics

- Gene Ontology
- pathway analysis
- enrichment
- GSEA
- over-representation analysis
- KEGG
- Reactome

Data Science connection

Statistical hypothesis testing + feature interpretation.

Practical

Analyze biological pathways associated with differentially expressed genes.

Lecture 20 — Network Biology

Topics

- biological networks
- protein-protein interaction networks
- gene regulatory networks
- co-expression networks
- centrality
- community detection
- graph representation

Tools

- Cytoscape
- STRING
- NetworkX

Data Science connection

Graph theory + graph machine learning.

MODULE 6 — SINGLE-CELL & SPATIAL BIOINFORMATICS

Lecture 21 — Single-Cell RNA Sequencing

Topics

- bulk vs single-cell RNA-seq
- cell heterogeneity

- single-cell workflow
- count matrix
- quality control
- normalization

Tools

- Seurat
- Scanpy
- AnnData

Lecture 22 — scRNA-seq Data Science

Topics

- highly variable genes
- PCA
- nearest-neighbor graphs
- UMAP
- t-SNE
- clustering
- cell-type annotation

Data Science connection

High-dimensional matrix → PCA → Graph → Clustering → UMAP

Practical

Perform an end-to-end Scanpy analysis.

Lecture 23 — Advanced Single-Cell Analysis

Topics

- trajectory inference
- pseudotime
- cell-cell communication
- differential abundance
- batch correction
- integration

Tools

- scVI
- Scanpy
- Seurat
- CellChat concepts

Lecture 24 — Spatial Transcriptomics & Spatial Omics

Topics

- what is spatial transcriptomics?
- spatial gene expression
- imaging-based methods
- spatial coordinates
- tissue architecture
- spatial domains
- cell-cell interactions

Modern concepts

- spatial transcriptomics
- spatial proteomics
- multimodal spatial omics
- spatial foundation models

Practical

Analyze and visualize a spatial transcriptomics dataset.

MODULE 7 — PROTEIN BIOINFORMATICS & AI

Lecture 25 — Protein Bioinformatics

Topics

- protein sequences
- amino acids
- domains
- motifs
- protein families
- protein function
- protein structure
- structure-function relationship

Databases

- UniProt
- PDB
- AlphaFold DB

AlphaFold DB currently provides access to more than 200 million predicted protein structures.

Lecture 26 — AI-Based Protein Structure Prediction

Topics

- AlphaFold 1
- AlphaFold 2
- AlphaFold 3
- protein language models
- structure prediction
- confidence scores
- pLDDT
- PAE
- molecular interactions

AlphaFold 3 extends prediction beyond isolated protein structures to complexes involving proteins, DNA/RNA, small molecules and other components.

Practical

Students retrieve and examine a protein prediction from AlphaFold DB.

Lecture 27 — Machine Learning for Bioinformatics

Topics

Supervised learning

- classification
- regression
- random forest
- SVM
- XGBoost

Unsupervised learning

- k-means
- hierarchical clustering
- PCA
- UMAP

Applications

- disease prediction
- biomarker discovery
- gene classification
- variant prediction
- drug response prediction

Practical

Build a disease-classification model using gene-expression features.

Lecture 28 — Deep Learning & Foundation Models for Biology

Topics

- CNNs for DNA sequences
- RNNs
- Transformers
- attention
- protein language models
- DNA language models
- embeddings
- transfer learning
- foundation models

Emerging concepts

- biological foundation models
- genomic transformers
- protein language models
- multimodal biological models
- AI agents for bioinformatics

The course should emphasize that AI outputs are **predictions requiring biological validation**, rather than automatically accepted scientific conclusions.

MODULE 8 — MULTI-OMICS, WORKFLOWS & RESEARCH

Lecture 29 — Multi-Omics Data Integration

Topics

- genomics
- transcriptomics
- proteomics
- metabolomics
- epigenomics
- multi-omics integration

Methods

- early integration
- late integration
- matrix factorization

- dimensionality reduction
- similarity networks
- representation learning

Practical

Integrate gene-expression and clinical features.

Lecture 30 — Reproducible Bioinformatics & Workflow Engineering

Topics

- reproducibility
- version control
- Git
- GitHub
- environments
- Conda/Mamba
- containers
- Docker
- Apptainer
- workflow managers

Modern workflow systems

- Nextflow
- nf-core
- Snakemake
- WDL

GATK's current best-practice ecosystem, for example, includes reference workflow implementations and WDL-based pipeline approaches to improve reproducibility.

Practical

Build a small reproducible pipeline:

FASTQ → QC → Alignment → Statistics

Lecture 31 — Cloud, HPC & Large-Scale Bioinformatics

Topics

- CPU vs GPU
- HPC clusters
- parallel processing
- cloud computing
- storage
- containers

- batch processing
- workflow scaling

Platforms

- Google Colab
- AWS
- Azure
- Google Cloud
- Terra
- HPC clusters

Data Science focus

Students learn how bioinformatics moves from:

Laptop → GPU → HPC → Cloud

Lecture 32 — Capstone Project & Future of Bioinformatics

Part A — Student Project Presentations

Each student/team presents an end-to-end project.

Part B — Future Trends

Discuss:

- AI-driven drug discovery
- genomic foundation models
- protein foundation models
- spatial foundation models
- digital biology
- multimodal biological AI
- AI agents for bioinformatics
- automated scientific workflows
- precision medicine
- population genomics
- pangenomes
- quantum computing possibilities
- ethical AI
- privacy-preserving genomics

The rapidly changing nature of the field should itself become a learning objective: students should learn how to evaluate whether a tool is **current, validated, reproducible and appropriate**, rather than simply memorizing software names.

